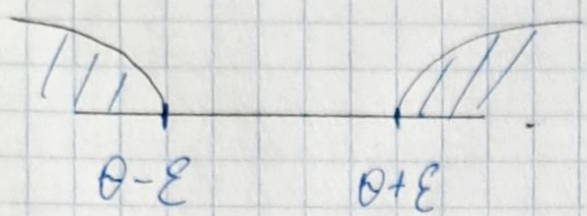
Т1

Не сделали на семинаре:

до-ть: 1) по опред. $\tilde{\theta}_3' = \frac{n+1}{n} x_{max}$ – состоятельна?

2) $\tilde{\theta}_3 = x_{max}$ – состоятельна?

2) $\tilde{\theta}_3$ – состоятельна, если $\forall \theta > 0 \ \forall \varepsilon > 0 \ P(|\tilde{\theta}_3 - \theta| \geq \varepsilon) \xrightarrow[n\to\infty]{} 0$



, т.к. $\xi \sim R(0;\theta) \Rightarrow P(x_{max} \geq \theta + \varepsilon) = 0$

$$P(|x_{max} - \theta| \geq \varepsilon) = P(x_{max} \leq \theta - \varepsilon) = \Phi(\theta - \varepsilon) = \left(\frac{\theta - \varepsilon}{\theta}\right)^n =$$

$$\left\{ \begin{array}{l} \xi \sim F(x) \Rightarrow \xi_{max} \sim (F(x))^n = \Phi(x) \\ \varphi(x) = \Phi'(x) = n\left(\frac{x}{\theta}\right)^{n-1} \frac{1}{\theta} \{(0;\theta)\} \end{array} \right\}$$

$$= \left(1 - \frac{\varepsilon}{\theta}\right)^n \xrightarrow[n\to\infty]{} 0 \Rightarrow \tilde{\theta}_3 \text{ – состоятельна}$$

1) $P(|\tilde{\theta}_3' - \theta| \geq \varepsilon) \leq \frac{D\tilde{\theta}_3'}{\varepsilon^2}$ – по нер-ву Чебышева

$$\frac{D\tilde{\theta}_3'}{\varepsilon^2} = \frac{\theta^2}{n(n+2)} \cdot \frac{1}{\varepsilon^2} \xrightarrow[n\to\infty]{} 0 \Rightarrow P(|\tilde{\theta}_3' - \theta| \geq \varepsilon) \xrightarrow[n\to\infty]{} 0 \Rightarrow$$

$\Rightarrow \tilde{\theta}_3'$ – состоятельна по определению.